

SANJAY SUTHRAYE

DATA SCIENTIST / AI/ML ENGINEER [AVAILABLE IMMEDIATELY] |  Portfolio

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SUMMARY

Data scientist with 7 years under my belt, I've been designing and putting into action AI/ML and core data science solutions in big environments. I have a solid grasp of generative AI, LLMs, and RAG systems, and I'm well-versed in NLP, predictive analytics, and machine learning architectures like LSTM, XGBoost, Random Forest, and CatBoost. I'm comfortable building intricate enrichment models, predictive systems, and time-series forecasting solutions, using the latest methods. I've also got a track record of leading cross-functional teams, all while developing end-to-end ML workflows.

SKILLS

Machine Learning & AI	Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Natural Language Processing (NLP), Machine Learning (XGBoost, Random Forest, Clustering), Predictive Modeling, Reinforcement Learning, Survival Analysis, Time-Series Forecasting, Model Explainability (XAI), Fairness & Bias Detection
ML Framework	Churn Prediction, Market Mix Modelling, Customer Segmentation, Enrichment Models
Libraries	PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers, LangChain, XGBoost, LightGBM, spaCy, NLTK, OpenCV, Pandas, NumPy
Cloud & MLOps	AWS (Lambda, ECS, EKS, S3, API Gateway, Serverless), GCP Vertex AI, Docker, Kubernetes, MLflow, Apache Airflow, Databricks, CI/CD Pipelines, Model Deployment, Model Monitoring
Programming & Tools	Python, R, SQL, PySpark, Spark, Snowflake, Git, Shell Scripting, REST APIs, Postman, Jupyter, VS Code
Data Engineering	Apache Kafka, BigQuery, MongoDB, Tableau, Power BI, Seaborn, Matplotlib, Dash
Leadership	Technical Mentorship, AI/ML Strategy, Cross-Functional Collaboration, Agile/Scrum, Stakeholder Communication, Ethical & Responsible AI

WORK EXPERIENCE

ASTRAZENECA

Senior Consultant - AI Engineer

Sep'25 - Present

Consultant - Data Scientist

Apr'23 - Sep'25

- Won the **Achiever Z Annual Award** (Sep'24) for my Patient Journey LLM project in the Lead AI and Insights category.
- Presented my Patient Journey LLM work at the **PMSA Conference** in New Delhi, receiving top recognition.
- Built an AI Marketing Agent platform with 8 modules, using **Langgraph**, deploying via REST API and WebSocket, which increased data access speed by **30%**.
- Developed multi-stage predictive models for Endometrial Cancer detection, improving diagnostic accuracy by **22%**.
- Created time series models (**LSTM**, **Sarimax**, **Prophet**) to forecast NBRx trends, cutting operational costs by **30%**.
- Designed a custom **RAG framework** for oncology that mapped disease-drug relationships through **Knowledge Graphs** and implemented guardrails, achieving 99% reliability in patient journey analysis.
- Created a UI for patient journey insights using **Neo4j**, **Open Cypher**, and **advanced prompt engineering**, cutting analysis time by **40%**.
- Built **NLP+RAG pipelines** with **HuggingFace**, **LangChain** and **BERT** that improved risk stratification by **20%** and sped up clinical decision-making.
- Combined EMR, claims, and diagnostic datasets into a single platform, providing 360° patient views and

increasing predictive accuracy by **25%**.

- Applied **causal inference** methods alongside **survival analysis** in retrieval models to develop precision medicine approaches.
- Built a multi-point enrichment model for Bladder Cancer with **0.7 F1 score** and performed care gap analysis.
- Wrote 40+ Python functions for statistical analysis with **CI/CD** integration, reducing analysis time by **65%**.

TESCO BENGALURU

Senior Applied Data Scientist

Aug'21 - Apr'23

- Received **Employee of the Month** award (Apr'22) through the Spotlight program for data science contributions.
- Led a **hackathon-winning project** (Apr'22) focused on employee health that improved workplace wellbeing.
- Developed a customer churn prediction model forecasting attrition 6 weeks ahead for **200K+** households using **Random Forest**, **Logistic Regression**, and **CatBoost**.
- Ran A/B tests on churn prevention models that increased household retention by **38%** and highlighted critical gaps in shopping patterns.
- Built a Product Choice Optimizer using a multi-model approach to maximize ROI, incorporating **Ad-stock variables** as marketing impact features.
- Created a Next-Gen Model that segmented high-value customers into 4 cohorts using **Kmeans++** and **DBScan**.

CARTESIAN CONSULTING

Senior Data Analyst

Feb'21 - Aug'21

- Created an ML model with **Levenshtein Algorithm** to find similar products, using the Pareto principle to focus on the top 20% of recommendations.
- Built the **Dealer Health Index** from scratch, handling everything from data collection to visualization in **Tableau** and Excel.
- Improved metrics for **62%** of underperforming dealers through data-driven recommendations.
- Used **text analysis** and similarity scoring to optimize product portfolios and boost sales efficiency.

DXC TECHNOLOGY

Senior Analyst at Helpdesk

Mar'19 - Apr'20

Technical Analyst (Contractor)

Jun'18 - Mar'19

- Designed an ML model using **Named-entity recognition** to analyze customer reviews, identifying key sentiment drivers.
- Optimized team traffic workflows with **regression analysis**, cutting average handling time by **15%**.
- Aligned helpdesk operations with industry standards through data-driven workflow improvements.
- Managed **SQL/MS SQL** databases and maintained a central knowledge base that improved information sharing.
- Analyzed call flow data to identify patterns that increased first-call resolution rates.

EDUCATION

University of Hyderabad

Jan'21 - Dec'22

Post Graduate Diploma in **Artificial Intelligence and Machine Learning**

GPA: 3.4/4.0

AMC College of Engineering

Jan'14 - Dec'18

Bachelor of Engineering

CGPA: 7.0/10.0

LANGUAGES

English - Fluent

German - A1 and A2 (Learning)